

Decimal ABM Enterprise Platform

A self-contained, zero-external-dependency Account-Based Marketing platform — HubSpot, Mailchimp, Apollo, Customer.io, Instantly, Smartlead and n8n rebuilt as native modules.

Master blueprint for a 26-module, zero-external-dependency ABM platform: HubSpot, Mailchimp, Apollo, Customer.io, Instantly, Smartlead and n8n rebuilt as native modules. Companion to the markdown spec repository

[ABM Enterprise Platform/](#).

Compiled 16 July 2026 · 26 modules · 21 domain folders

SECTION 1

Vision — Zero Human Intervention

The objective is Zero Human Intervention. External SaaS (HubSpot, Mailchimp, Apollo, Clay, Instantly, Smartlead, n8n) must stop being dependencies and become **native modules** the platform owns end-to-end. Instead of Signal Engine -> Clay -> Apollo -> HubSpot -> Mailchimp -> Smartlead, the flow becomes a single owned pipeline: Signal Engine -> ABM Intelligence Layer -> ABM CRM Engine (HubSpot replica) -> ABM Marketing Engine (Mailchimp replica) -> ABM Outreach Engine -> ABM Analytics Engine. This document is the complete engineering blueprint: exhaustive enough that an AI-assisted team (Claude Code / Cursor / Codex) can build it in ~5 days, with enterprise-grade quality, multi-tenancy, RBAC, Arabic/English i18n, and full extensibility.

From dependency chain to owned pipeline

BEFORE (RENTED)

```
Signal Engine
-> Clay
-> Apollo
-> HubSpot
-> Mailchimp
-> Smartlead
```

AFTER (OWNED)

```
Signal Engine
-> ABM Intelligence Layer
-> ABM CRM Engine (HubSpot replica)
-> ABM Marketing Engine (Mailchimp replica)
-> ABM Outreach Engine
-> ABM Analytics Engine
```

How zero-intervention is achieved. Zero Human Intervention is achieved by the Rules Engine (15) + Workflow Engine (16) + Journey Engine (08) driving the pipeline autonomously, the AI Engine (10) generating content, and the autonomy ladder (A1-A5) governing how much runs without review — with three hard stops that always require a human: (1) C-suite outreach, (2) anything failing compliance/consent/suppression, (3) actions above the earned trust tier. The Copilot (26) is the human's window in; the platform runs without it.

SECTION 2

The 26 modules at a glance

#	MODULE	DOMAIN FOLDER	REPLACES
01	Intelligence Engine	02_Intelligence	The 'brain'
02	Signal Detection Engine	03_Signal_Detection	6sense/Bombora intent + Google Alerts + custom scrapers
03	Contact & Account Enrichment Engine	04_Contact_Enrichment	Apollo + Clay + ZoomInfo + Lusha
04	Contact Intelligence Engine	04_Contact_Enrichment	HubSpot Contacts + Apollo person records
05	Account Engine	05_Account_Management	HubSpot Companies + 6sense account model
06	CRM Engine (HubSpot Replica)	06_CRM_Engine	HubSpot CRM in full
07	Marketing Automation Engine (Mailchimp Replica)	07_Marketing_Automation	Mailchimp + Customer.io + Brevo
08	Journey Builder Engine	07_Marketing_Automation	HubSpot Workflows + Customer.io journeys
09	Campaign Builder Engine	07_Marketing_Automation	HubSpot Campaigns + orchestration wrapper over ABM plays.
10	AI Personalization Engine	10_AI_Engine	Jasper/Copy.ai/Clay AI columns + custom prompt stacks
11	Email Delivery Engine	08_Email_Engine	Mandrill/SendGrid MTA + Postfix
12	LinkedIn Automation Engine	09_LinkedIn_Engine	Smartlead/Expandi/Dux-Soup
13	Landing Page & Forms Engine	07_Marketing_Automation	HubSpot Landing Pages + Forms + Unbounce + popups/preference center.
14	Asset Library	07_Marketing_Automation	HubSpot Files + DAM
15	Rules Engine	13_Rules_Engine	HubSpot workflow logic + custom IFTTT
16	Workflow Engine (n8n-style)	11_Workflow_Engine	n8n / Zapier / Make
17	Analytics Engine	12_Analytics	HubSpot Analytics + Mailchimp Reports + product analytics

18	Lead & Account Scoring Engine	05_Account_Management	HubSpot scoring + 6sense/MadKudu
19	Pipeline Management Engine	05_Account_Management	HubSpot deal pipelines + forecasting.
20	Reporting Engine	12_Analytics	HubSpot reports/dashboards + exports + scheduled digests.
21	Notification Engine	14_Admin	HubSpot notifications + Slack/Teams alerts.
22	Attribution Engine	12_Analytics	HubSpot attribution + Bizible-style multi-touch models.
23	API Gateway	15_API	Kong/Apigee + HubSpot public API surface.
24	Integration Layer & Event Bus	16_UI	Internal Kafka/Redis event bus + connector framework (optional external bridges).
25	Admin Console & User/Permission Management	14_Admin	HubSpot settings + super-admin + billing/quotas + RBAC.
26	AI Copilot	10_AI_Engine	HubSpot ChatSpot / Breeze

SECTION 3

Repository structure & technology stack

The 20-folder spec repository

FOLDER	CONTENTS
00_MASTER/	Master architecture, global data model, event catalog, cross-cutting specs, 5-day build plan, glossary.
01_Core_Platform/	Platform foundation: multi-tenancy, config, base entities, shared kernel, service/repository patterns.
02_Intelligence/	Module 01 Intelligence Engine + EPIS confidence spine + reasoning streams.
03_Signal_Detection/	Module 02 Signal Detection Engine (8 capture sub-streams, filter, decay).
04_Contact_Enrichment/	Module 03 Enrichment Engine + Module 04 Contact Intelligence Engine.
05_Account_Management/	Module 05 Account Engine + Module 18 Lead/Account Scoring + Module 19 Pipeline.
06_CRM_Engine/	Module 06 CRM Engine (full HubSpot replica).
07_Marketing_Automation/	Modules 07 Marketing, 08 Journey, 09 Campaign, 13 Landing/Forms, 14 Asset Library (Mailchimp+ replica).
08_Email_Engine/	Module 11 Email Delivery Engine (MTA, event pipeline, deliverability).
09_LinkedIn_Engine/	Module 12 LinkedIn Automation Engine (ban-risk gated).
10_AI_Engine/	Modules 10 AI Personalization + 26 AI Copilot.
11_Workflow_Engine/	Module 16 Workflow Engine (n8n-style node automation).
12_Analytics/	Modules 17 Analytics + 20 Reporting + 22 Attribution.
13_Rules_Engine/	Module 15 Rules Engine (no-code IF/THEN core).
14_Admin/	Modules 21 Notification + 25 Admin/User/Permission Management.
15_API/	Module 23 API Gateway.
16_UI/	Front-end app: dashboards, builders (journey/workflow/email/page), CRM UI, copilot; + Module 24 Integration Layer/event bus specs.
17_Database/	Global schema, ER diagrams, migrations strategy, partitioning, materialized views.
18_Deployment/	Docker/compose, environments, secrets, scaling, observability, backups, DR.
19_Testing/	Test strategy, fixtures, contract tests, load/perf, acceptance suites.

Standard document set per module

Functional Specification (FSD), Business Requirements (BRD), Technical Design (TDD), Database Design & ER, API Design, Business Rules, State Machines, Sequence Diagrams, Activity Diagrams, Events, Workflows, Permissions & RBAC, Acceptance Criteria, Edge Cases, Testing, Future Enhancements.

Technology stack

LAYER	CHOICE
Language / runtime	Python 3.12+ (async), TypeScript (front-end)
API framework	FastAPI + Pydantic v2 (async, OpenAPI-native)
ORM / migrations	SQLAlchemy 2.x + Alembic
Primary DB	PostgreSQL 16 — UUID PKs, JSONB, partitioning, materialized views, row-level security
Cache / queue / bus	Redis (cache + Streams event bus) or Kafka for scale; Celery/RQ workers
Search	PostgreSQL FTS + optional OpenSearch for cross-object search
Vector / embeddings	pgvector for semantic search + document embeddings
AI models	Provider-agnostic adapter (Gemini / Anthropic / OpenAI) via Integration Layer; PII anonymized
Email MTA	Pluggable: Mandrill-style API adapter + SMTP; native event pipeline (pixel/redirect/webhooks)
Front-end	React + Tailwind; visual builders for journey/workflow/email/landing
Auth	OAuth2 / JWT + API keys; RBAC deny-by-default; multi-tenant row-level security
Infra	Docker + docker-compose (dev) / Kubernetes (scale); public HTTPS ingress for webhooks
Observability	Structured logging, metrics, tracing, health endpoints, audit logs
i18n	English + Arabic (RTL), locale-aware sending & content

SECTION 4

Global data model & event architecture

Shared kernel entities

ENTITY	DESCRIPTION	OWNER
<code>tenant</code>	Workspace/org — root of multi-tenancy.	Admin (25)
<code>user / team / role</code>	Identity + RBAC.	Admin (25)
<code>account</code>	Target organization (bank/fintech/subsidiary/vendor).	Account Engine (05)
<code>contact</code>	Person at an account.	Contact Engine (04)
<code>company</code>	Non-target org / vendor / partner.	CRM (06)
<code>relationship</code>	Graph edge (org/person/vendor/tech).	CRM (06) / Graph
<code>signal / raw_capture</code>	Captured intelligence + provenance.	Signal Engine (02)
<code>intelligence_record / nba / hypothesis</code>	Reasoned intelligence.	Intelligence (01)
<code>deal / pipeline / stage</code>	Revenue objects.	CRM (06) / Pipeline (19)
<code>activity</code>	Universal interaction record.	CRM (06)
<code>campaign / journey / enrollment</code>	Orchestration objects.	Campaign (09)/ Journey (08)
<code>email_campaign / message / delivery_event</code>	Marketing + delivery.	Marketing (07)/ Delivery (11)
<code>account_score / lead_score / modifier</code>	Scoring.	Scoring (18)
<code>event</code>	Platform event (bus).	Integration Layer (24)
<code>audit_log</code>	Change history.	CRM (06)/Admin (25)

Event bus catalog (representative)

EVENT	PRODUCER	CONSUMERS
<code>signal.created</code>	Signal Engine	Intelligence, Scoring, Account, Enrichment
<code>signal.cluster.promoted</code>	Signal Engine	Intelligence

<code>intelligence.record.created</code>	Intelligence	Scoring, CRM, Copilot
<code>intelligence.nba.created</code>	Intelligence	CRM, Notification, Copilot
<code>enrichment.entity.updated</code>	Enrichment	Contact, Account, CRM, Scoring
<code>contact.consent.changed</code>	Contact	Marketing, Journey, Delivery
<code>account.tiered</code>	Account	Enrichment, Journey, Notification
<code>account.held</code>	Account	Journey, LinkedIn, Marketing (pause cascade)
<code>score.updated</code> / <code>score.threshold.crossed</code>	Scoring	Account, Intelligence, Analytics, Notification
<code>deal.stage.changed</code>	CRM/ Pipeline	Pipeline, Attribution, Analytics, Marketing (transactional)
<code>email.campaign.sent</code>	Marketing	Delivery, Analytics
<code>email.event.opened/clicked/bounced/complained</code>	Delivery	Marketing, Contact, Scoring, Attribution, Analytics
<code>email.reply.received</code>	Delivery	Account (pause), Journey (pause), Notification
<code>journey.enrolled</code> / <code>step.executed</code> / <code>goal.met</code> / <code>exited</code>	Journey	Analytics, Campaign, Attribution
<code>linkedin.reply.received</code> / <code>seat.cooldown</code> / <code>circuit_breaker.tripped</code>	LinkedIn	Account, Journey, Notification
<code>form.submitted</code> / <code>consent.captured</code>	Landing/ Forms	Contact, Journey, Analytics
<code>rule.fired</code>	Rules	Audit, target engines
<code>workflow.run.finished</code> / <code>approval.requested</code>	Workflow	Notification, Analytics
<code>meeting.booked</code>	Calendar/ CRM	Account, Pipeline, Scoring, Attribution, Notification
<code>quota.exhausted</code>	Admin	Notification, calling engine (block)

At-least-once delivery · per-key ordering · idempotent consumers · schema registry · retry → DLQ → replay. Engines stay decoupled through the bus (Module 24).

SECTION 5

Cross-cutting concerns (1/2)

Multi-tenancy

Every entity carries `tenant_id`; row-level security enforces isolation; no cross-tenant reads. Tenant provisioning, plans, suspension in Admin (25).

AuthN / AuthZ / RBAC

OAuth2/JWT + API keys at the Gateway (23). RBAC is deny-by-default, permissions additive via roles (25). Row-level security ties records to owner/team; managers see team, admins see tenant.

Audit logging

Every privileged mutation writes an audit entry (actor, before/after, timestamp). CRM (06) and Admin (25) own the audit stores; all engines emit.

Event architecture

Async event bus (24) is the nervous system: at-least-once delivery, per-key ordering, idempotent consumers, schema registry + versioning, retry/DLQ/replay. Engines are decoupled producers/consumers.

Error handling & retries

Standard error envelope (code, message, detail, `trace_id`). Transient failures retry with backoff; exhausted -> DLQ or failure path. Outreach failures never silently drop.

Scheduling & workers

Central scheduler emits `schedule.tick`; background workers run captures, scoring, rollups, journey advancement, sends. PID/lease locks prevent double-run.

Queue architecture

Per-domain queues (send, enrichment, ai, workflow) with priority + rate control; back-pressure via throttling.

Caching

Redis for hot reads (scores, segments, dashboards); cache invalidation on the owning entity's change event.

SECTION 5

Cross-cutting concerns (2/2)

Search architecture

Postgres FTS for in-app search; cross-object search (CRM) optionally OpenSearch; pgvector for semantic/document search.

Performance & scalability

Partition large tables (contacts, activities, events, messages); materialized views for analytics; horizontal worker scaling; 1M+ contacts and high send volume as design targets.

Security

Secrets via vault refs (never inline); signed webhooks; PII anonymization before LLM; encryption at rest/in transit; least-privilege service accounts; PDPL compliance.

Internationalization

English + Arabic (RTL) across UI and generated content; locale-aware timezones; KSA calendar (Sun-Thu, Ramadan blackout) enforced in sending.

Compliance spine

Consent (none/opted_in/denied) + do_not_contact + suppression enforced at enroll AND send; HMAC-signed unsubscribe; account-centric pause; C-suite always human review; PDPL legal gate before live outreach.

Configuration management

Per-tenant settings + feature flags in Admin; environment config via env vars; no hardcoded secrets.

Deployment architecture

Containerized services behind API Gateway; public HTTPS ingress for inbound webhooks (email/LinkedIn/calendar); managed Postgres + Redis; blue/green deploys.

Testing strategy

Unit + contract (OpenAPI) + integration (event flows) + property (idempotency/ordering) + load + acceptance suites per module; golden-file tests for scoring & reasoning determinism.

Microservice boundaries

Each numbered engine is an independently deployable service (or a bounded module in a modular monolith for the 5-day build, split later). Services never share tables across boundaries; they communicate via the event bus (24) and the API Gateway (23). The CRM (06) owns the relational spine; other engines reference CRM objects by id and subscribe to CRM events. Scoring (18), Intelligence (01), and Analytics (17) are read-heavy consumers. Delivery (11), LinkedIn (12), Workflow (16) are execution services. Admin (25) + Gateway (23) + Integration Layer (24) are the platform control/plumbing plane.

01–03 · Intelligence Engine → Contact & Account Enrichment Engine

01 · Intelligence Engine

Replaces: The 'brain' — no direct external equivalent; the orchestration layer above Clay/Apollo/6sense intent.

Purpose. The central reasoning and orchestration layer that turns raw captured signals and enriched graph data into ranked, explained, action-ready intelligence. It owns the confidence model (EPIS), the reasoning streams (HYP/VSAT/POWER/TENSION/MOBILITY), and the 'why now / who / what to say' synthesis every downstream engine consumes.

Signal reasoning streams and hypothesis generation

EPIS confidence calibration on every derived fact

Account/opportunity synthesis (why-now narratives)

Next-Best-Action computation feeding CRM & Journey engines

Intelligence briefs (account, persona, meeting)

Key entities: intelligence_record, hypothesis, nba_recommendation, evidence_ref.

Signature APIs: GET /v1/intelligence/accounts/{id}/brief; GET /v1/intelligence/accounts/{id}/nba; POST /v1/intelligence/hypotheses:generate; POST /v1/intelligence/records:query.

Top rules: INT-001 INT-002 INT-003 INT-004 — full set in the module spec.

Publishes: intelligence.record.created, intelligence.nba.created, intelligence.hypothesis.confirmed

02 · Signal Detection Engine

Replaces: 6sense/Bombora intent + Google Alerts + custom scrapers — capture & first-pass classification.

Purpose. Autonomous capture of buying and context signals across eight sourcing sub-streams, with provenance, idempotency, relevance filtering, dedup, and decay — producing clean, deduped, confidence-eligible signals for the Intelligence Engine.

8 capture sub-streams (NEWS/REG/EXEC/VENDOR/SUBS/EVENT/FIN/PATH)

raw_captures provenance log + dedup

SIG-RELEVANCE 4-axis filter

Signal decay stamping

Clustering related signals

Key entities: raw_capture, signal, signal_cluster, source.

Signature APIs: GET /v1/signals; POST /v1/signals; POST /v1/signals/{id}:reclassify; GET /v1/sources.

Top rules: SIG-001 SIG-002 SIG-003 SIG-004 — full set in the module spec.

Publishes: signal.created, signal.cluster.promoted, signal.filtered_out, source.ban_risk.tripped

03 · Contact & Account Enrichment Engine

Replaces: Apollo + Clay + ZoomInfo + Lusha — data acquisition, waterfall enrichment, verification.

Purpose. Acquire, resolve, verify and continuously refresh contact and company data into the graph — a native waterfall enrichment pipeline with provider adapters, email/phone verification, and identity resolution, so the platform owns its data instead of renting Apollo.

Provider adapter framework (pluggable data sources)

Waterfall enrichment (try sources in priority order)

Email/phone verification & scoring

Identity resolution & merge

Refresh scheduling & staleness detection

Key entities: enrichment_job, provider, verification, merge_candidate.

Signature APIs: POST /v1/enrichment/jobs; GET /v1/enrichment/jobs/{id}; POST /v1/enrichment/verify; GET /v1/enrichment/merge-candidates.

Top rules: ENR-001 ENR-002 ENR-003 ENR-004 — full set in the module spec.

Publishes: enrichment.entity.updated, enrichment.verification.completed, enrichment.merge.detected

04–06 · Contact Intelligence Engine → CRM Engine

04 · Contact Intelligence Engine

Replaces: HubSpot Contacts + Apollo person records — the people system of record & scoring.

Purpose. System of record for people at scale (architected for 1M+ contacts): profile, career history, persona classification, decision authority, engagement history, and per-contact intelligence — the authoritative contact object every other engine references.

Contact CRUD & bulk import

Persona & seniority classification

Decision authority / buying influence scoring

Engagement history rollup

Consent & suppression state on the person

Key entities: contact, career_event, contact_engagement.

Signature APIs: GET /v1/contacts; POST /v1/contacts; PATCH /v1/contacts/{id}; POST /v1/contacts:bulk.

Top rules: CON-001 CON-002 CON-003 CON-004 — full set in the module spec.

Publishes: contact.created, contact.updated, contact.classified, contact.consent.changed

05 · Account Engine

Replaces: HubSpot Companies + 6sense account model — the account system of record & tiering.

Purpose. System of record for target organizations and their structure (parents, subsidiaries, vendors, tech stack), the account scoring inputs, tiering (HOT/WARM/COLD), and the account-centric orchestration rule that one reply pauses the whole account.

Account CRUD & hierarchy (parent/subsidiary)

Tech stack & vendor mapping on the account

Account tier assignment & daily budget

Account-level pause/hold state

Product-fit mapping (account x product)

Key entities: account, account_tech, product_fit, account_hold.

Signature APIs: GET /v1/accounts; POST /v1/accounts; PATCH /v1/accounts/{id}; GET /v1/accounts/{id}/graph.

Top rules: ACC-001 ACC-002 ACC-003 ACC-004 — full set in the module spec.

Publishes: account.created, account.tiered, account.held, account.released

06 · CRM Engine (HubSpot Replica)

Replaces: HubSpot CRM in full — accounts, contacts, deals, activities, custom objects, properties, timelines, workflows.

Purpose. A full native CRM: the relational spine tying accounts, contacts, companies, buying committees, relationship graph, deals, pipelines, activities, tasks, notes, custom objects and properties into one auditable system of record with AI timeline, engagement scoring, next-best-action, forecasting, duplicate/merge, search, views, lists and segments — reproducing HubSpot CRM capability natively.

Objects: Account, Contact, Company, Deal, Opportunity, Activity, Task, Note, Meeting, Call, Email, Custom Object

Buying Committee & Relationship Graph

Properties framework (custom fields) + Lead Status + Lifecycle

Pipelines & stages (thin ref to Pipeline Engine 19)

Owners/Teams/Permissions, Views/Lists/Segments/Tags

Key entities: `crm_object`, `property`, `deal`, `activity`, `committee_member`, `relationship`, `view`, `crm_task`, `audit_log`.

Signature APIs: `GET /v1/crm/{object}/{id}`; `POST /v1/crm/{object}`; `PATCH /v1/crm/{object}/{id}`; `POST /v1/crm/properties`.

Top rules: CRM-001 CRM-002 CRM-003 CRM-004 — full set in the module spec.

Publishes: `crm.contact.changed`, `crm.deal.changed`, `deal.stage.changed`, `crm.merged`

07–09 · Marketing Automation Engine → Campaign Builder Engine

07 · Marketing Automation Engine (Mailchimp Replica)

Replaces: Mailchimp + Customer.io + Brevo — audience, segmentation, templates, sending, tracking, automation, deliverability.

Purpose. A full native marketing-automation system: audiences & dynamic segments, suppression lists, template/email builder with AI generation, sending with open/click/bounce tracking, A/B & multivariate testing, personalization/merge tags/dynamic blocks, transactional + drip automation, deliverability (IP warming, domain/DKIM/SPF/DMARC, SMTP), scheduling with timezone/send-time optimization — replacing Mailchimp entirely.

Audiences, Lists, Segments (static+dynamic), Suppression lists

Template Builder / Drag-drop Email Builder / AI Email + Subject generator

Preview, Spam checker, Link/Open/Click/Bounce tracking, Heatmaps

A/B + Multivariate testing

Personalization: merge tags, dynamic blocks

Key entities: audience, suppression, template, email_campaign, email_message, ab_test, domain.

Signature APIs: POST /v1/mkt/audiences; GET /v1/mkt/audiences/{id}/members; POST /v1/mkt/templates; POST /v1/mkt/templates/generate.

Top rules: MKT-001 MKT-002 MKT-003 MKT-004 — full set in the module spec.

Publishes: email.campaign.sent, email.event.opened, email.event.clicked, email.event.bounced

08 · Journey Builder Engine

Replaces: HubSpot Workflows + Customer.io journeys — visual multi-step, multi-channel orchestration.

Purpose. Visual, multi-channel journey orchestration: an account/contact enters a journey and moves through steps, conditions, triggers, actions, delays, goals and exit conditions across email, LinkedIn, WhatsApp, tasks and webhooks — the drip/sequence brain that replaces Smartlead/Instantly sequencing natively, with pause-on-reply and the account-centric rule enforced.

Journey canvas: steps, branches, delays, waits

Enrollment (from segment/list/trigger/rule)

Conditions & decision splits

Multi-channel actions (email/LinkedIn/WhatsApp/task/webhook/AI)

Goals & exit conditions

Key entities: journey, journey_step, enrollment, journey_event.

Signature APIs: POST /v1/journeys; POST /v1/journeys/{id}:activate; POST /v1/journeys/{id}:enroll; GET /v1/journeys/{id}/analytics.

Top rules: JRN-001 JRN-002 JRN-003 JRN-004 — full set in the module spec.

Publishes: journey.enrolled, journey.step.executed, journey.goal.met, journey.exited

09 · Campaign Builder Engine

Replaces: HubSpot Campaigns + orchestration wrapper over ABM plays.

Purpose. The strategic wrapper that groups journeys, email campaigns, LinkedIn sequences, landing pages, assets and target account lists into a single named ABM campaign with objectives, budget, timeline, membership and unified ROI — the object leadership plans and reports against.

Campaign object: objective, audience (account list), budget, timeline, owner

Membership: journeys, email campaigns, sequences, assets, landing pages under one campaign

Unified campaign analytics & ROI/pipeline attribution rollup

Campaign templates (reusable ABM plays)

Key entities: campaign, campaign_member, campaign_metric.

Signature APIs: POST /v1/campaigns; POST /v1/campaigns/{id}/members; GET /v1/campaigns/{id}/roi; POST /v1/campaigns:from-template.

Top rules: CMP-001 CMP-002 CMP-003 CMP-004 — full set in the module spec.

Publishes: campaign.activated, campaign.completed

10–12 · AI Personalization Engine → LinkedIn Automation Engine

10 · AI Personalization Engine

Replaces: Jasper/Copy.ai/Clay AI columns + custom prompt stacks — content generation & personalization.

Purpose. The content-generation brain: turns intelligence (brief, persona, pain, signals) into channel-ready, on-brand, compliant copy — emails, subjects, LinkedIn messages, landing-page copy, proposals, case studies, meeting prep — via a governed prompt/orchestration layer with the 7-agent chain, QC guardrails, and PII anonymization.

7-agent chain: Signal Analyst->Account Research->Persona Psychology->Pain Inference->Strategy->Message Gen->QC

Generators: email, subject, LinkedIn, landing page, proposal, case study, meeting prep, call summary

Prompt Builder & template library

Brand voice + teaser-discipline QC guardrails

PII anonymization (no real names/emails to LLM)

Key entities: generation, prompt, brand_voice, prediction.

Signature APIs: POST /v1/ai/generate; POST /v1/ai/generate:bulk; POST /v1/ai/qc; POST /v1/ai/prompts.

Top rules: AIP-001 AIP-002 AIP-003 AIP-004 — full set in the module spec.

Publishes: ai.generation.created, ai.generation.qc_failed, ai.prediction.updated

11 · Email Delivery Engine

Replaces: Mandrill/SendGrid MTA + Postfix — the actual send + event pipeline.

Purpose. The low-level send-and-track layer: accept a rendered message, deliver it via MTA/provider with IP-pool + reputation management, and capture the full event pipeline (delivered/open/click/bounce/complaint) that everything upstream depends on — the native replacement for Mailchimp's real moat.

Send API (single + batch) with queueing & rate control

Provider/MTA adapters (SMTP, Mandrill-style API) + failover

Open pixel, click redirect, bounce/complaint webhooks

Feedback-loop registration, suppression sync

IP pool & warmup enforcement, throttling

Key entities: send_request, delivery_event, ip_pool, provider_route.

Signature APIs: POST /v1/delivery/send; POST /v1/delivery/send:batch; POST /v1/delivery/webhooks/{provider}; GET /v1/delivery/messages/{id}/events.

Top rules: DEL-001 DEL-002 DEL-003 DEL-004 — full set in the module spec.

Publishes: email.event.delivered, email.event.opened, email.event.clicked, email.event.bounced

12 · LinkedIn Automation Engine

Replaces: Smartlead/Expandi/Dux-Soup — LinkedIn touches, gated by ban-risk controls.

Purpose. Native, safety-first LinkedIn outreach: connection requests, messages, InMail, profile views and post engagement as journey/sequence steps — behind a strict ban-risk circuit breaker, human-like pacing, and per-seat daily limits. Deliberately the last capability activated.

LinkedIn action steps: connect, message, InMail, view, follow, like

Per-seat daily limits + human-like randomized pacing

Ban-risk circuit breaker + anomaly halt

Seat/session management (per-user auth)

Reply detection -> pause-on-reply

Key entities: li_seat, li_action, li_sequence.

Signature APIs: POST /v1/linkedin/seats; POST /v1/linkedin/actions; GET /v1/linkedin/seats/{id}/health;
POST /v1/linkedin/circuit-breaker:status.

Top rules: LI-001 LI-002 LI-003 LI-004 — full set in the module spec.

Publishes: linkedin.action.sent, linkedin.reply.received, linkedin.seat.cooldown,
linkedin.circuit_breaker.tripped

13–15 · Landing Page & Forms Engine → Rules Engine

13 · Landing Page & Forms Engine

Replaces: HubSpot Landing Pages + Forms + Unbounce + popups/preference center.

Purpose. Native landing pages, forms, popups, preference & unsubscribe centers: a builder + hosting + submission pipeline that captures leads directly into the CRM, powers gated assets (whitepaper/case-study downloads), and manages consent — closing the loop the BFSI whitepaper campaign needed.

Landing page builder (blocks) + hosting + SEO/OG/JSON-LD

Forms (fields, validation, progressive profiling, hidden UTM)

Popups & preference center + unsubscribe center

Submission -> contact create/update + journey enroll + consent capture

Gated content delivery (asset unlock)

Key entities: landing_page, form, submission, preference.

Signature APIs: POST /v1/pages; POST /v1/forms; POST /v1/forms/{id}/submit; GET /p/{slug}.

Top rules: LP-001 LP-002 LP-003 LP-004 — full set in the module spec.

Publishes: form.submitted, page.published, preference.updated, consent.captured

14 · Asset Library

Replaces: HubSpot Files + DAM — content & collateral store.

Purpose. Central store for marketing/sales collateral (whitepapers, case studies, one-pagers, decks, images) with versioning, gating, usage tracking and CDN delivery — the source of truth for every asset a campaign, landing page or email references.

Asset upload + versioning + metadata/tags

Gating flag + signed/expiring links

Usage tracking (downloads, which campaigns use it)

CDN/hosting + access control

AI-generated asset intake (proposals/case studies from AI Engine)

Key entities: asset, asset_usage.

Signature APIs: POST /v1/assets; GET /v1/assets; POST /v1/assets/{id}:sign; GET /v1/assets/{id}/usage.

Top rules: AST-001 AST-002 AST-003 AST-004 — full set in the module spec.

Publishes: asset.published, asset.downloaded

15 · Rules Engine

Replaces: HubSpot workflow logic + custom IFTTT — the no-code IF/THEN decision core.

Purpose. The configurable no-code decision core every enterprise product revolves around: WHEN/IF (conditions over signals, scores, entities, events) THEN (actions across any engine). Non-developers compose rules like 'IF signal_score>80 AND funding>\$50M AND CTO exists THEN create opportunity, assign owner, generate brief+email+LinkedIn sequence, start campaign, wait 3 days, check open...' — evaluated deterministically and auditable.

Rule authoring (conditions + actions) no-code

Condition operators over any entity/field/event/score

Action catalog (create/assign/generate/enroll/notify/update/wait/branch)

Rule evaluation engine (event-driven + scheduled)

Rule versioning, priority, conflict resolution, dry-run/simulate

Key entities: rule, rule_firing, action_def, condition_op.

Signature APIs: POST /v1/rules; POST /v1/rules/{id}:simulate; POST /v1/rules/{id}:activate; GET /v1/rules/{id}/firings.

Top rules: RUL-001 RUL-002 RUL-003 RUL-004 — full set in the module spec.

Publishes: rule.fired, rule.action.executed, rule.conflict.detected

16–18 · Workflow Engine → Lead & Account Scoring Engine

16 · Workflow Engine (n8n-style)

Replaces: n8n / Zapier / Make — visual node-based automation inside the platform.

Purpose. A visual, node-based automation builder (like n8n but native): drag nodes — Email, LinkedIn, Webhook, CRM, Condition, Delay, Wait, Decision, Loop, Merge, Split, HTTP, Python, LLM, News/RSS, Slack, Teams, WhatsApp, SMS, Call, Meeting, Calendar, Approval, Manual Step, AI Step — wire them into durable, resumable workflows that power everything from onboarding to complex multi-branch plays.

Visual DAG builder with 25+ node types

Durable execution (resumable across waits/restarts)

Triggers: event, schedule, webhook, manual

Data mapping between nodes; expressions

Error handling, retries, timeouts per node

Key entities: workflow, workflow_run, node_execution, credential.

Signature APIs: POST /v1/workflows; POST /v1/workflows/{id}:run; GET /v1/workflows/runs/{id}; POST /v1/workflows/runs/{id}:resume.

Top rules: WFL-001 WFL-002 WFL-003 WFL-004 — full set in the module spec.

Publishes: workflow.run.started, workflow.run.finished, workflow.node.failed, workflow.approval.requested

17 · Analytics Engine

Replaces: HubSpot Analytics + Mailchimp Reports + product analytics — metrics & funnels.

Purpose. Unified analytics across accounts, campaigns, journeys, pipeline, revenue, email, LinkedIn, AI performance and workflows — an event-sourced metrics layer with funnels, cohorts, conversion, response/meeting rates and dashboards, feeding both the Reporting Engine and the Copilot.

Event ingestion from the platform event bus into an analytics store

Metric definitions & aggregations (rollups, funnels, cohorts)

Domain analytics: account, campaign, journey, pipeline, revenue, email, LinkedIn, AI, workflow

Conversion funnels, response/meeting rates, CAC hooks

Query API + materialized dashboards

Key entities: metric_event, metric_def, rollout, funnel.

Signature APIs: POST /v1/analytics/query; GET /v1/analytics/funnels/{id}; POST /v1/analytics/metrics; GET /v1/analytics/dashboards/{key}.

Top rules: ANL-001 ANL-002 ANL-003 ANL-004 — full set in the module spec.

Publishes: analytics.rollup.completed, analytics.anomaly.detected

18 · Lead & Account Scoring Engine

Replaces: HubSpot scoring + 6sense/MadKudu — the scoring math & Effective-Opportunity model.

Purpose. The quantitative core: computes the 0-100 account score (Signal 35% / Regulatory 30% / Reachability 20% / Relationship 15%), the Effective-Opportunity equation with its modifier chain, and contact-level lead scores — recalculated daily and on material events, driving tiering, routing and prioritization.

Account base score (4 weighted dimensions)

Effective-Opportunity = $\text{Dynamic_Score} \times \text{ICS}/100 \times \text{Stage} \times \text{Budget} \times \text{Entrenchment} \times \text{Risk} \times \text{Window}$

Contact lead scoring (fit + engagement)

Modifier lookup table (Bible Artifact 1)

Daily recompute + event-driven recompute

Key entities: account_score, lead_score, modifier, score_event.

Signature APIs: GET /v1/scoring/accounts/{id}; POST /v1/scoring/accounts/{id}:recompute; GET /v1/scoring/accounts/{id}/history; PUT /v1/scoring/modifiers.

Top rules: SCO-001 SCO-002 SCO-003 SCO-004 — full set in the module spec.

Publishes: score.updated, score.threshold.crossed

19–21 · Pipeline Management Engine → Notification Engine

19 · Pipeline Management Engine

Replaces: HubSpot deal pipelines + forecasting.

Purpose. Deal pipeline configuration and progression: multiple pipelines, custom stages with entry/exit criteria and probabilities, weighted forecasting, gap-to-quota and pipeline-health analytics — the revenue spine the CRM deals move along.

Pipeline & stage config (per product/segment)

Stage entry/exit criteria & default probabilities

Weighted forecast (commit/likely/worst)

Pipeline health: stalled deals, single-threaded, hygiene

Quota & gap analysis

Key entities: pipeline, stage, forecast, pipeline_health.

Signature APIs: POST /v1/pipelines; GET /v1/pipelines/{id}/forecast; GET /v1/pipelines/{id}/health; PATCH /v1/pipelines/{id}/stages.

Top rules: PIP-001 PIP-002 PIP-003 PIP-004 — full set in the module spec.

Publishes: forecast.updated, pipeline.deal.stalled, pipeline.health.flagged

20 · Reporting Engine

Replaces: HubSpot reports/dashboards + exports + scheduled digests.

Purpose. Turns analytics into consumable, shareable output: custom report builder, prebuilt dashboards, scheduled email digests, exports (PDF/CSV/XLSX), and the executive-brief generator (one-click pre-meeting PDF) — the presentation layer over the Analytics Engine.

Report builder (pick metrics/dims/viz)

Dashboard composition (11 role-based dashboards)

Scheduled digests (daily/weekly)

Exports: PDF/CSV/XLSX

Executive Brief generator (account one-pager PDF)

Key entities: report, dashboard, schedule, brief.

Signature APIs: POST /v1/reports; GET /v1/reports/{id}/render; POST /v1/reports/{id}:export; POST /v1/reports/schedules.

Top rules: REP-001 REP-002 REP-003 REP-004 — full set in the module spec.

Publishes: report.exported, digest.sent, brief.generated

21 · Notification Engine

Replaces: HubSpot notifications + Slack/Teams alerts.

Purpose. Central multi-channel notification and alerting: in-app, email, Slack, Teams, WhatsApp — delivering NBAs, replies, hot-account alerts, approvals, anomalies and digests to the right person with preferences, batching and escalation.

Notification templates + channels (in-app/email/Slack/Teams/WhatsApp)

User notification preferences & quiet hours

Real-time alerts (reply, hot account, approval needed, anomaly)

Batching/digest + escalation policies

Delivery tracking of notifications

Key entities: notification, notify_pref, escalation.

Signature APIs: POST /v1/notifications:send; GET /v1/notifications; POST /v1/notifications/{id}:read; PUT /v1/notifications/preferences.

Top rules: NOT-001 NOT-002 NOT-003 NOT-004 — full set in the module spec.

Publishes: notification.sent, notification.escalated

22–24 · Attribution Engine → Integration Layer & Event Bus

22 · Attribution Engine

Replaces: HubSpot attribution + Bizible-style multi-touch models.

Purpose. Assigns credit for pipeline and revenue across the touches, campaigns, journeys and channels that influenced an account — multi-touch attribution models (first/last/linear/time-decay/W-shaped/custom) that turn the activity graph into ROI the Campaign and Reporting engines consume.

Touch capture into attribution paths

Attribution models: first, last, linear, time-decay, U/W-shaped, custom

Account-based attribution (credit at account not just contact)

Campaign/channel/journey ROI rollup

Model comparison & configurable window

Key entities: touch, attribution_path, attribution_result.

Signature APIs: GET /v1/attribution/campaigns/{id}; POST /v1/attribution:recompute; GET /v1/attribution/models:compare.

Top rules: ATT-001 ATT-002 ATT-003 ATT-004 — full set in the module spec.

Publishes: attribution.recomputed

23 · API Gateway

Replaces: Kong/Apigee + HubSpot public API surface.

Purpose. The single, secured entry point for all external and inter-service API traffic: routing, authentication, authorization, rate limiting, quotas, versioning, request/response validation, API keys, webhooks-out and developer docs — the front door to the whole platform.

Request routing to services

AuthN (OAuth2/JWT/API key) + AuthZ (RBAC/scopes)

Rate limiting, quotas, throttling per tenant/key

API versioning + deprecation

Request/response validation (OpenAPI)

Key entities: api_key, route, webhook_out, rate_bucket.

Signature APIs: POST /v1/admin/api-keys; GET /v1/meta/openapi; POST /v1/admin/webhooks; GET /v1/admin/usage.

Top rules: GW-001 GW-002 GW-003 GW-004 — full set in the module spec.

Publishes: api.key.created, api.webhook.failed, api.ratelimit.exceeded

24 · Integration Layer & Event Bus

Replaces: Internal Kafka/Redis event bus + connector framework (optional external bridges).

Purpose. The nervous system: the internal event bus that lets every engine publish/subscribe asynchronously, plus a connector framework for optional external bridges (Slack, Teams, WhatsApp, calendar, model providers) — the plumbing that makes 26 modules one platform while keeping them decoupled.

Internal event bus (publish/subscribe, durable, ordered per key)

Event schema registry + versioning

Connector framework for optional externals (Slack/Teams/WhatsApp/Calendar/LLM providers)

Retry, dead-letter, replay

Idempotency & exactly-once-ish delivery semantics

Key entities: event, subscription, dead_letter, connector.

Signature APIs: POST /v1/events:publish; POST /v1/subscriptions; GET /v1/events/dlq; POST /v1/events/dlq/{id}:replay.

Top rules: INT-BUS-001 INT-BUS-002 INT-BUS-003 INT-BUS-004 — full set in the module spec.

Publishes: bus.dlq.added, connector.status.changed

25–26 · Admin Console & User/Permission Management → AI Copilot

25 · Admin Console & User/Permission Management

Replaces: HubSpot settings + super-admin + billing/quotas + RBAC.

Purpose. The control plane: organizations/tenants, teams, users, roles & permissions (RBAC), API keys, domains, SMTP, integrations, branding, audit logs, usage, billing, quotas, AI credits, monitoring and health — everything an operator configures and governs, with full multi-tenancy.

Tenant/org management (multi-tenancy)

Users, teams, roles, granular permissions (RBAC)

API keys, domains, SMTP, integrations, branding

Usage, billing, quotas, AI credits, rate limits

Audit logs (platform-wide), monitoring, health, feature flags

Key entities: tenant, user, team, role, quota, audit_entry.

Signature APIs: POST /v1/admin/tenants; POST /v1/admin/users:invite; POST /v1/admin/roles; GET /v1/admin/usage.

Top rules: ADM-001 ADM-002 ADM-003 ADM-004 — full set in the module spec.

Publishes: tenant.provisioned, user.invited, role.changed, quota.exhausted

26 · AI Copilot

Replaces: HubSpot ChatSpot / Breeze — conversational interface over the whole platform.

Purpose. The natural-language interface to everything: 'Which banks should I call today?', 'How do I approach Riyadh Bank?', 'Draft outreach to the AI Rajhi CDO', 'What changed on my accounts this week?' — a governed agentic copilot that queries the graph, invokes engines (with permission + compliance gates), and returns answers, briefs and actions. The realization of the zero-human-intervention vision as an assistant surface.

Conversational NL query over graph/analytics

Action invocation (create task, enroll, generate draft, move deal) via tool-calling

Grounded answers with citations to intelligence/records

Guardrails: permission-aware, compliance-gated, confidence-qualified

Proactive suggestions (today's priorities, risks)

Key entities: copilot_session, copilot_turn, tool_binding.

Signature APIs: POST /v1/copilot/ask; GET /v1/copilot/sessions/{id}; GET /v1/copilot/suggestions; GET /v1/copilot/tools.

Top rules: COP-001 COP-002 COP-003 COP-004 — full set in the module spec.

Publishes: copilot.action.taken, copilot.query.answered

SECTION 7

5-day AI-assisted build plan

Day 1 — Foundation

Core Platform (01): multi-tenancy, base entities, shared kernel, config, RBAC scaffolding. Database (17): global schema + migrations. Integration Layer (24): event bus + schema registry. API Gateway (23) skeleton. Admin (25) tenants/users/roles. Outcome: a multi-tenant, event-driven skeleton with auth.

Day 2 — Data & Intelligence

Signal Engine (02): raw_capture, SIG-NEWS worker, filter, decay. Enrichment (03) waterfall + verify. Contact (04) + Account (05) + Scoring (18): the account-first model + 0-100 score + Effective-Opportunity. Intelligence (01) + EPIS + reasoning streams. Outcome: signals flow to scored, tiered accounts with briefs.

Day 3 — CRM & Marketing replicas

CRM Engine (06): objects, properties, deals, activities, committee, graph, views, merge, audit, timeline. Marketing (07) + Journey (08) + Campaign (09): audiences, templates, campaigns, journeys. Email Delivery (11): send + event pipeline. Outcome: HubSpot + Mailchimp replicas operational end-to-end.

Day 4 — Automation & AI

Rules Engine (15): no-code IF/THEN. Workflow Engine (16): node automation. AI Personalization (10): 7-agent chain + QC. Landing/Forms (13) + Assets (14). LinkedIn (12) behind circuit breaker. Notification (21). Outcome: autonomous orchestration with AI content + compliance gates.

Day 5 — Insight, Copilot, Hardening

Analytics (17) + Attribution (22) + Reporting (20) + exec briefs. AI Copilot (26). Pipeline (19) forecasting. Cross-cutting hardening: RBAC matrix, quotas, i18n, KSA calendar, load tests, acceptance suites. Outcome: full platform with copilot, analytics, and the feedback loop closed.

Guardrails: never delete production data; always migrate; inspect before modifying; compliance gates built in from Day 1; every module ships with its acceptance suite.

REFERENCE

How the blueprint & repo fit together

This blueprint is the spine. The companion repository `ABM_Enterprise_Platform/` contains 26 module specifications across 21 domain folders, each following an 18-section template (purpose, scope, personas, entities/schema, API contracts, workflows, state machine, events, business rules, RBAC, validations, errors, integrations, testing, acceptance criteria, edge cases, implementation checklist, future).

An AI-assisted engineering team should build folder-by-folder following the 5-day plan, using each module's implementation checklist and acceptance criteria as the definition of done.

This redesign turns the ABM engine from an execution playbook that depends on HubSpot/Mailchimp/Apollo/Smartlead/n8n into a single enterprise platform that owns every layer natively — the precondition for true zero-human-intervention operation.